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Underground Mining and Bulk Materials Handling (MIN E – 630) Assignment #1

1 question 1

To get the length of the decline, which means the distance between the the bottom and the portal of the decline in 3D, so we can use 3D distance formula to calculate it.

$$L = \sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2} \quad (1)$$

Where: L is the length of the decline; Δx is the horizontal distance in the east-west direction; Δy is the horizontal distance in the north-south direction; Δz is the vertical distance

Apply the values provided into the Equation 1:

$$L = \sqrt{(6400 - 7000)^2 + (10500 - 9600)^2 + (4260 - 4010)^2} \approx 1110.18 \text{ ft}$$

So the length of the decline is approximately 1110.18 ft.

2 question 2

Since the top of ore body thickness and the bottom of ore body thickness are different, we assume the ore body thickness changes linearly from the top to the bottom. So the average thickness of the ore body can be calculated as:

$$Thickness_{avg} = \frac{10 + 20}{2} = 15 \text{ m}$$

To get the ore deposit based on the given data, we can use the formula below:

$$\text{Ore Deposit} = \text{Volume} \times \text{Density} \quad (2)$$

For volume,

$$\begin{aligned} \text{Volume} &= \text{Area} \times Thickness_{avg} \\ &= \text{Strike distance} \times \text{Dip distance} \times Thickness_{avg} \\ &= 400 \times 500 \times 15 \\ &= 30000000 \text{ m}^3 \end{aligned}$$

Therefore, apply the volume and density into the Equation 2:

$$\text{Ore Deposit} = 30000000 \times 2.45 = 73500000 \text{ t}$$

3 question 3

As shown in the Figure 1.

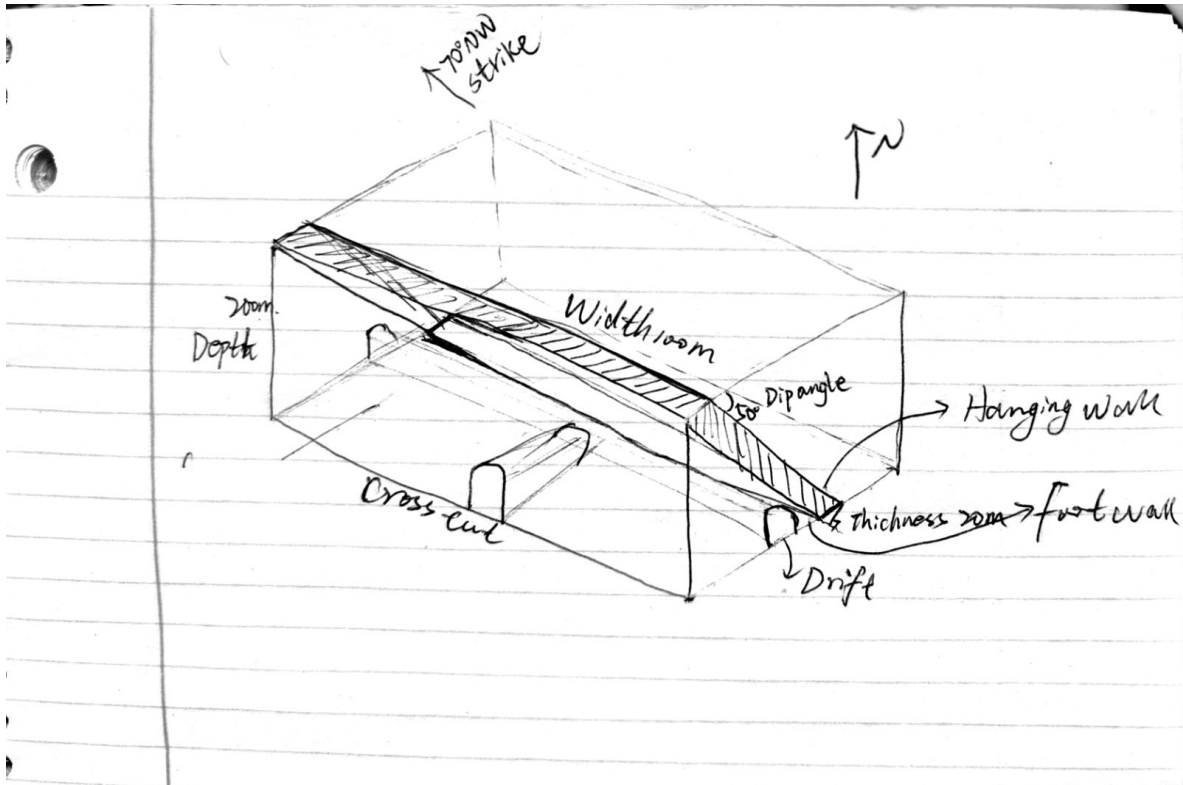


Figure 1: a freehand isometric sketch for this vein

4 question 4

4.1 subquestion 1

The process to solve Rock Substance Strength (RSS) is shown in the below.

Per method selection by David E. Nicholas,

$$RSS = \frac{UCS}{\text{Overburden pressure}}$$

Overburden pressure = $\rho g H$, ρ = specific gravity $\times \rho_{\text{water}}$

Apply given data, we have:

Item	UCS	Overburden pressure	RSS
Hanging wall	91 MPa	$1000 \times 2.7 \times 9.8 \times 1200$ Pa	2.866
Ore	103 MPa	$1000 \times 2.7 \times 9.8 \times 1200$ Pa	3.244
Foot wall	125 MPa	$1000 \times 2.7 \times 9.8 \times 1200$ Pa	3.937

4.2 subquestion 2

The snapshot of UofA Macdonald Mining Method Selection Tool is shown in the Figure 2.

Macdonald Mining Method Selection - University of Alberta - Department of Civil and Environmental Engineering

This tool was created for the University of Alberta Dean's Research Award under supervision of Wei Victor Liu, published in 2023. The following is based off the paper: Nicholas, D.E. 1981. Method selection-A numerical approach. In Design and Operation of Caving and Sublevel Stopping Mines. Edited by D. Stewart. New York: SME-AIME. The tool allows for ease of calculation of a ranked list of mining methods from the geological properties of a deposit and the depiction of the orebody from given parameters. [Some underlined text contains tooltips for further explanation on definitions](#)

Inputs

Orebody Geometry and Grade Distribution

Shape:

Thickness:

Plunge:

Grade Distribution:

Rock Mass Rating (RMR)

	Rock Substance Strength	Fracture Spacing	Fracture Strength
Ore	Weak (<8)	Close (10-16/m or 20-30% RQD)	Moderate
Hanging Wall	Weak (<8)	Wide (3-10/m or 40-70% RQD)	Moderate
Footwall	Weak (<8)	Very Wide (<3/m or 70-100% RQD)	Moderate

Rankings and Visualization

Rank	Score	Method
1	42	Open-Pit
2	38	Cut-and-Fill Stopping
3	36	Block Caving
4	30	Sublevel Caving
5	30	Shrinkage Stopping
6	-29	Room-and-Pillar
7	-66	Longwall Mining
8	-77	Sublevel Stopping

Figure 2: Screenshot of Macdonald Mining Method Selection

Based on the result of the UofA Macdonald Mining Method Selection Tool, the first suitable method is open pit, but we can't choose it due to the deposit depth up to 1200m, so the first two suitable underground methods are:

1. Cut-and-Fill Method scores 38, ranks 1st in underground method.
2. Block Caving Method scores 36, ranks 2nd in underground method.